

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 4

INDIAN SPACE RESEARCH ORGANISATION - 2

CRYOGENIC ROCKET ENGINE

- A cryogenic rocket engine uses a cryogenic fuel and oxidizer.
- Cryogenic fuels are fuels that require storage at extremely low temperatures in order to maintain them in a liquid state.
- The fuel and oxidizer are both gases which are liquefied and stored at very low temperatures. Cryogenic fuels are mostly liquefied gases such as liquid hydrogen.
- Various cryogenic fuel-oxidizer combinations have been tried but the combination of liquid hydrogen and liquid oxygen is the most widely used.
- Oxygen liquefies at -183 degrees Celsius and hydrogen liquefies at -253 degrees Celsius.



CRYOGENIC ROCKET ENGINE

- Cryogenic engine is a very complex system, due to the use of propellants at extremely low temperature. It also requires complex ground support systems like propellant storage and filling systems, transportation and handling of cryogenic fluids and testing facilities.

- **Benefits**

- ✓ Cryogenic fuels are environmentally cleaner than other rocket fuels.
- ✓ Cryogenic fuels have a higher mass flow rate than fossil fuels and therefore, produce more thrust and power when combusted in an engine. This means that engines will run farther on less fuel.
- ✓ Cryogenic fuels are non-pollutants and therefore, if spilled, pose no risk to the environment. There will be no need to clean up hazardous waste after a spill.



2. GSLV- GEOSYNCHRONOUS LAUNCH VEHICLE

- Geosynchronous Satellite Launch Vehicle-GSLV Mk II, is an expendable launch vehicle operated by ISRO. GSLV was launched first in 2001.
- The GSLV project was initiated in 1990 with the objective of acquiring a launch capability for geosynchronous satellites.
- GSLV is a three-stage vehicle with solid, liquid and cryogenic stages respectively.





- **First Stage: GS1**
 - ✓ The first stage uses a solid rocket motor and is augmented by 4 liquid strap-on boosters.
- **Second Stage: GS2**
 - ✓ Vikas engine (liquid propellant) is used in the second stage of GSLV.
- **Third Stage: Cryogenic Upper Stage (CUS)**
 - ✓ The third stage uses indigenously developed Cryogenic Engine known as CE-7.5, developed under the Cryogenic Upper Stage Project (CUSP). The CE-7.5 is India's first cryogenic engine, developed by the Liquid Propulsion Systems (LPSC).

The Vikas Engine

- *The Vikas Engine (a portmanteau from initials of VIKram Ambalal Sarabhai) is a family of liquid fueled rocket engines developed by the Liquid Propulsion Systems Centre in the 1970s.*
- *Vikas Engine is the workhorse liquid rocket engine powering the second stage of India's Polar Satellite Launch Vehicle (PSLV), second stage and the four strap on boosters of Geosynchronous Launch Vehicle (GSLV) and the twin engine core liquid stage of GSLV Mk-III.*

GSLV VARIANTS

- GSLV rockets using the Russian Cryogenic Stage (CS) are designated as the GSLV Mk I while versions using the indigenous Cryogenic Upper Stage (CUS) are designated the GSLV Mk II.
- **GSAT-9/The South Asia Satellite**
- It is also known as the SAARC Satellite.
- It is a geostationary communications and meteorology satellite operated by ISRO for the SAARC nations except Pakistan. The satellite was launched on 5th May 2017.
- During the 18th SAARC summit held in Nepal in 2014, the Indian PM proposed the idea of a satellite serving the needs of SAARC nations.
- Afghanistan, Bangladesh, Bhutan, Maldives, Nepal and Sri Lanka are the users of this satellite.
- It was launched by the GSLV Mk II F09 vehicle.



WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

0866 244 44 72

98499 42299

94415 27588